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Detecting Health Misinformation: AI Solutions for Medical and Policy Content

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***Abstract*— The dissemination of health misinformation in a short period is life-threatening to the trusting attitudes, medical decision-making, and adherence to policies. In the given research, AI-based methods of automated classifying health-related claims are offered using transformer-based language models and hybrid structures into Supports, Refutes, and Neutral categories. Various state-of-the-art biomedical NLP models have been tested such as BioBERT, RoBERTa, SciBERT, ClinicalBERT, and a fine-tuned Meta-Llama-3-8B-Instruct. A lot of experiments on the HealthVer dataset prove that the Meta-Llama-3-8B-Instruct has a maximum performance score, with its own test accuracy of 0.8391 and the Macro-F1 score of 0.7831, which is even better than BioBERT and other models. It was found in comparative analysis that medical domain-specific transformers are of great use in credibility assessment in clinical and public-health claims. Such results indicate the possibility of sophisticated large-language model to be used as an effective instrument in fighting health misinformation on digital pillars.**

Keywords*— *Health misinformation detection, Natural Language Processing, Transformer models, LLMs, Biomedical text classification, Fact-checking.

I. INTRODUCTION

As a branch of artificial intelligence, Natural Language processing (NLP) aims to help machines comprehend, read, and reason natural languages. As the volume of digital textual information grows exponentially, NLP is now a core technology in automation of wide scale information processing, decision support, and knowledge discovery in many areas. In recent years, transformer architecture models and deep learning have greatly enhanced the capacity of NLP systems to understand semantic context, low-resource settings, as well as broaden their generalizations to language structure diversity.

Consequently, the use of NLP has been successfully applied in several real-world domains, including news analysis [1], intent classification [2], agricultural decision making [3], and legal document comprehension [4], [5]. These developments underscore the capability of NLP in handling domain-specific language and complex reasoning tasks, highlighting its potential to address critical societal challenges. The healthcare domain is one such high-stakes area where accurate and reliable language understanding is essential.

The blistering rise of online communication tools has radically disrupted the nature of accessing and consuming medical information by people. Online forums, social media, and news portals help to propagate health-related claims at an unprecedented rate, especially in the case of the global crisis in the COVID-19 pandemic. Though this accessibility has the positive effect of enhancing the level of public awareness, it equally spurs the spread of false information in shape of false, misleading or scientifically unproven claims. The consequences of these inaccuracies can impact the medical decision-making process negatively, decrease vaccine acceptance, encourage unsafe treatment administration, and undermine the health policies of society. Therefore, health misinformation is turning out to be a significant hazard around the world and a situation requiring immediate scalable and automated solutions capable of determining the veracity of medical and policy-related words or utterances on a real-time basis.

The conventional systems of fact-checking highly depend on manual confirmations by the experts that are time-consuming, expensive and cannot match the speed at which the online content is created. The progress in both NLP and deep learning offers a possibility to make the process of misinformation detection automated, as now the models can see and assess the situation in biomedicine, interpret evidence, and decide on whether the health statement is true or not. Incorporating misinformation detection as a problem of stance classification, there are three possible categories of claims: supported, refuted, and neutral, which are structured according to the scientific evidence they provide. Nevertheless, this task is not easy as it is dependent on domain specific terminology, class disbalance, small semantic differences between categories of evidence, and scarcity of high quality annotated bio medical datasets.

Recent advances in transformer based architectures and large language models (LLMs) have strongly indicated likely improvement of automated reasoning in complex and high stakes domains. Different methods to achieve better robustness and interpretability on a range of NLP tasks, such

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as verification, legal analysis, and intent understanding, have been applied using domain-specific pretraining, parameter-efficient fine-tuning, and instruction-based learning. Such advancements encourage the quest of how such models can be successfully used in the realm of healthcare misinformation detection where fine-grained reasoning and interpretation of evidence play a crucial role in making a correct classification.

On the basis of such advancements, this paper explores how health misinformation can be automatically detected with the use of state-of-the-art transformer-based models on a biomedical fact verification benchmark. The problem is specified as a stance classification problem, according to which medical and policy-related statements are defined as Support, Refute, or Neutral regarding available scientific evidence. The paper is aimed at overcoming the following challenges: class imbalance and semantic ambiguity in biomedical claims are addressed using data-level and model-level strategies, which will allow obtaining more reliable and scalable AI-assisted health misinformation detection.

The rest of the paper is structured in the following way. Section II provides a literature review of the relevant research in the fields of health misinformation detectors, biomedical fact-checkers, and transformer-based NLP systems. Section III outlines the dataset, preprocessing, data augmentation policy and the experimental approach that will be applied. Section IV gives out the experimental results and the discussion of the results. Lastly, Section V provides the conclusion of the paper and the potential research directions.

The significant contributions of this work are:

- Comparative analysis of transformer-based biomedical language models and an instruction-tuned large language model in the automated identification of health misinformation on the HealthVer dataset.
- An adaptive data augmentation approach based on the underrepresented Refute category to enhance misinformation detection.
- A study on the improvement in the input representation formats and their effects on the effectiveness of classifying

II. Related Works

This literature review examines the more recent studies that are specialized in automated health misinformation, evidence-based medical fact-checking, and domain-specific natural language processing systems. The chosen papers cover the establishment of benchmark datasets, transformer-based verifying models, explainable fact-checking models, and large language model based reasoning systems that are implemented on claims related to the medical field.

Sarrouti et al. [6] confirmed that in order to validate the validity of claims concerning health matters, one would need real-world medical data, not artificial datasets. They presented the HEALTHVER dataset that included 14,330 claim-evidence pairs that were developed in a three-step procedure that includes claim retrieval, computerized ranking of relevant scientific papers, and manual labeling of claim-evidence relationships with SUPPORT, REFUTE, and NEUTRAL. They have found that deep learning model trained on actual medical claims can recall and suggest ongoing medical falsehoods much better than other models trained on fake claims, thus showing that HEALTHVER can be used to offer a more natural SOA into the study of medical fact-checking in the future.

Vladika et al. [7] proposed HealthFC, a dataset, which can support evidence-based fact-checking of multilingual health claims using expert-verified evidence provided by clinical trials and systematic reviews. They indicated that the dataset has 750 health related claims in both German and English and has baseline outcomes in terms of fact-checking and explanation generating task. The authors concluded that HealthFC is a difficult resource in terms of the further development of automated fact-checking techniques in the health sphere and it can serve as a promising field of investigation in the future.

Kotonya and Toni [8] highlighted the lack of research to explainable fact-checking to domains that require special knowledge like in the case of public health. They presented the PUBHEALTH dataset of 11.8K claims with explanations of the claims written by journalists and explored veracity prediction and explanation generation problems. Their results demonstrated that domain-specific training is more effective in increasing explainable performance of fact-checking, and concluded that coherent explanations would be more viable and reliable in automated fact-checking systems.

Barve and Saini [9] reported about how misinformation can be identified and fact-checked on healthcare webpages with the help of machine learning. They suggested a model which extracts URLs, preprocesses content, computes the frequency of terms and uses sentimental and grammatical results and distance measures like euclidean, Jaccard, and cosine similarity to determine which claims are true and which are false. They found that Naive Bayes was the best in the accuracy of 98.7 percent and Jaccard distance gave better results than other similarity measures indicating the significance of sentiment features in detecting misinformation.

Deka et al. [10] and claimed that authentic and relevant medical evidence retrieval is an important element to measure the validity of health information in the online environment, especially in the COVID-19 pandemic. They suggested query-based retrieval and transformer-based filtering methods that were used unsupervised to retrieve reliable scientific information in scholarly repositories. As a conclusion, they have decided that their suggested techniques are effective at enhancing the retrieval of the relevant articles to qualify claims, and they have a great possibility to boost evidence-based detection of health misinformation.

Srba et al. [11] introduced the increasing role of medical misinformation and the Monant dataset of about 317k medical news articles and 3.5k medical news articles of claims that have been fact-checked with over 51k manually labeled mappings. They furnished grounds on claim presence detection and stance classification tasks and tested them with manually labeled ground truth. Their findings proved that the dataset can be used to conduct research on misinformation characterization and diffusion and provided an excellent source of fact-checking based on machine learning.

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Walter et al. [12] provided a meta-analysis study on fact-checking effectiveness in correcting false information in 30 studies with 20,963 participants. They claimed that fact-checking produces the positive corrective effect with an effect size of d = 0.29 but they pointed out that the effect faded on certain instances like the presentation of truth scale and partial refutations. They found that the success of the fact-checking process is strongly predetermined by preexisting beliefs and ideology, which discloses the essential gaps and areas of the future research.

Nakov et al. [13] intelligent searches to create surveyed powerful methods of assisting professional fact-checkers in recognizing and validating online claims. They surveyed the techniques to detect claims, retrieve evidence, identify stance, and predict veracity in addition to identifying the limitation in the amount of misinformation, finding credibility, and the usability of tools. They came to a conclusion that the use of automated tools along with human expertise could contribute substantially to the efficiency of fact-checking and applicability to the real world.

Bhatnagar et al. [14] presented the primary technique used by automated fact-checking systems is to retrieve a previously verified fact using keyword-based retrieval, although it does not work well with the noisy text of social media. Abstractive summarization has proven useful in creating brief text representations though it has not been extensively applied in claim retrieval. The gap in this work is filled by means of summarization, which enables the construction of more effective queries and the enhancement of the retrieval performance considerably.

Bayani et al. [15] introduced an automated fact-checking tool of online health information to PubMed as the evidence source, based on transformers. Their model categorized web page content under thematic topics, auto-generated queries and auto-retrieved pertinent literature and compared similarities with cosine similarity and Jaccard distance. Their experiments revealed that the best score of BERT models was 96 percent in terms of F1 and cosine similarity was better than Jaccard, and that PubMed-based verification is possible with excellent health information evaluation with the help of algorithms.

Various studies that have been conducted recently have dealt with automated fact-checking and detection of misinformation within the healthcare sector. The development of large-scale datasets, including HEALTHVER and HealthFC, provided one of the lines of research with an opportunity to validate real-world multilingual medical claims that are supported by scientific evidence. Other studies were directed at explainable fact-checking via resources like PUBHEALTH that offer gold-standard explanations to boost the interpretability. Misinformation detectors based on machine learning have shown high accuracy with sentiment feature-based models and approaches like similarity measure models recording accuracy rates as high as 98.7%. Further studies offered methods of sophisticated retrieval and refinement to acquire the pertinent evidence in the scholarly repositories and created great amounts of data connecting claims and stance-labeled articles to facilitate the study of misinformation. Fact-checking Studies based on meta-analysis stated a positive influence of corrective beliefs, which are moderated by ideology of the user and prior knowledge. Surveys of automated fact-checking pipelines prioritize on claim detection, retrieval of evidence, as well as verification. Better evidence and reasoning frameworks More recent models use big-language models to unify evidence with reasoning in order to be more reliable and less hallucinogenous. Online health information tools based on PubMed retrieval using transformers have demonstrated good performance in the facilitation of automated validation of online health information. On the whole, these articles show that medical fact-checking has already made major steps forward, at the same time, challenges such as explainability, evidence quality, and generalization are observed.

III. Methodology

This section outlines the general approach that is taken to automated health misinformation detection. The proposed method is based on several steps, such as data analysis, preprocessing, data augmentation to overcome the problem of class imbalance, input representation generation, training of models, and evaluation. The entire methodological pipeline would be summarized in a rough sketch in Fig. 1 that shows the sequential procedures of operations of the proposed system. All the elements of the methodology are described in the subsections below.

A. Dataset Description and Class Distribution

The entire experimentation in this paper was based on the HealthVer dataset, which is a benchmark corpus that is explicitly designed to verify evidence-based biomedical facts. The dataset comprises expert annotation of claim–evidence pairs which have been derived based on peer-reviewed medical literature. Each example has (i) a natural language medical statement, (ii) a relevant evidence passage corresponding to the statement, which has been retrieved from PubMed-indexed articles, and (iii) a stance label that represents the factual connection between the statement and the evidence. The stance labels are divided into three exclusive classes: Support, Refute, and Neutral. The preliminary statistical results of the raw HealthVer data indicated that the class imbalance was enormous, with the Refute class being significantly underrepresented in com parison to the Support and Neutral classes. The imbalance is especially worrisome in tasks of detecting misinformation because refuting claims can be accompanied by the most severe public health consequences and are more semantically difficult. Previous researchers have indicated that such dis torted distributions may favor learning algorithms toward the majority classes, resulting in low recall and F1-scores for minority classes. To vividly show such an imbalance, Fig. 2 shows the distribution of the samples per class in the original HealthVer dataset before any balancing or augmentation techniques were applied. The dominance of the Support and Neutral classes stands out in the visualization and presents the necessity of applying corrective actions, which include both algorithmic and data-level interventions.

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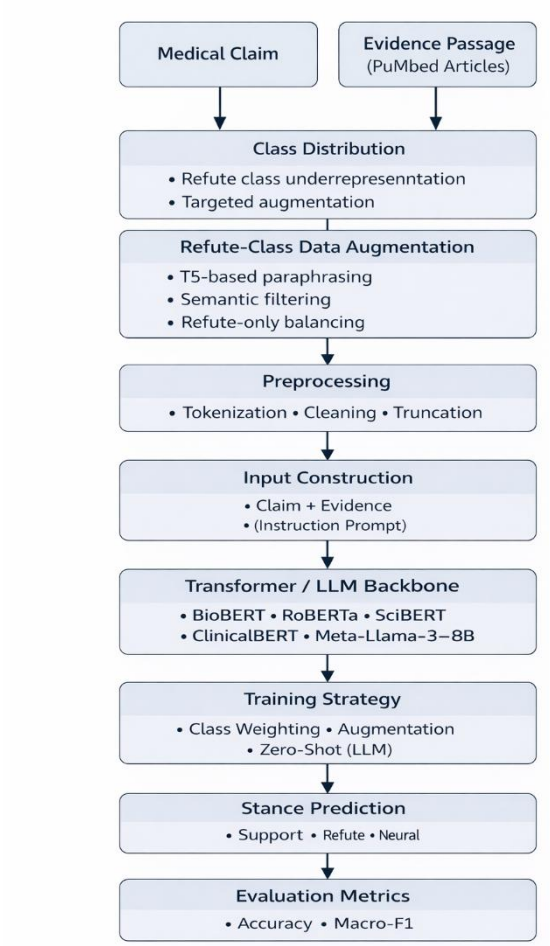


Fig. 1. General architecture for health misinformation detection using transformer-based models and large language models.

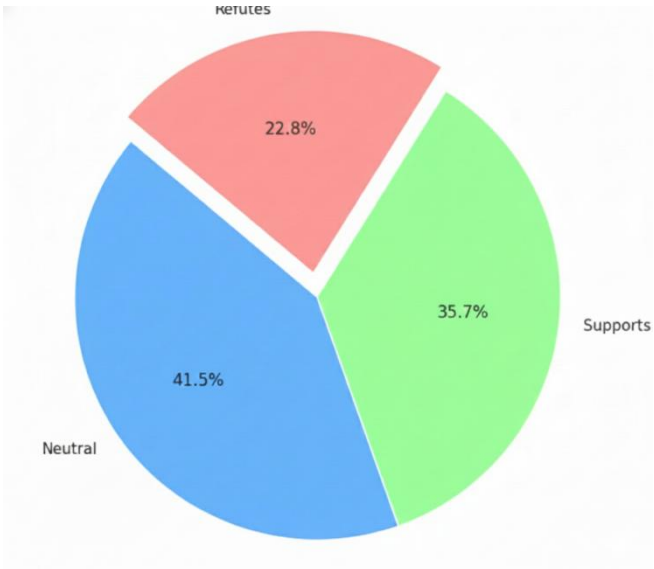


Fig. 2. Class distribution of the original HealthVer dataset (raw data).

B. Maintaining the Integrity of the Specifications

To overcome the problem of underrepresentation of the Refute class, a data augmentation strategy specifically oriented to address this issue was employed, based on a T5-based paraphrasing model. Augmentation was applied only to the Refute class in order to preserve the natural distribution and semantic properties of the Support and Neutral classes.

The claims labeled as Refute were paraphrased in a con trolled manner to maintain the original semantic content while ensuring lexical and syntactic diversity. To produce high quality paraphrases, beam search decoding was used, and out puts that deviated semantically or introduced ambiguity were discarded through manual inspection and heuristic filtering. Through this process, 1,089 additional Refute samples were generated, bringing the representation of this minority class to approximately match the other two classes.

The effect of this augmentation process can be observed in Fig. 3, which shows the class distributions before and after paraphrasing. The figure illustrates a substantially improved balance, enabling models to better learn refutation patterns that are crucial for detecting health misinformation.

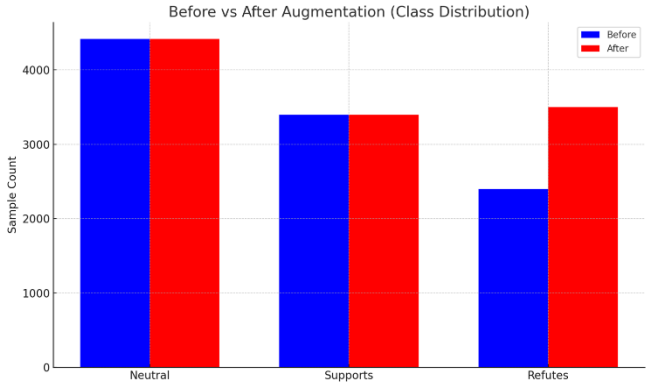


Fig. 3. Class distribution before and after T5-based paraphrasing augmenta tion.

C. Preprocessing

A unified and standardized preprocessing pipeline was ap plied across all models to ensure experimental consistency. This pipeline included whitespace normalization, removal of non-informative artifacts, and tokenization using model specific tokenizers (e.g., BioBERT tokenizer for BioBERT and RoBERTa tokenizer for RoBERTa). All input sequences were truncated or padded to a fixed maximum length based on model constraints to maintain uniform input dimensions.

A consistent numerical mapping was used for label encoding across all experiments (Support = 0, Refute = 1, Neutral = 2), ensuring compatibility with different learning architectures and facilitating comparative evaluation.

D. Input Representation Formats

Two different input representation formats were explored, with the objective of influencing stance classification performance.

1) *Two-Column Format (Baseline):* In the baseline setup, the claim and its corresponding evidence passage were concatenated into a single sequence using a special separator token. This format is consistent with conventional transformer-based sentence-pair classification approaches.
A representative example formatted under the two-column format is shown below:
Claim: Results on the use of hydroxychloroquine as a treatment for Covid-19 have shown no significant differences

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in health outcomes between the control group and patients who received the experimental drug.
Evidence: HCQ should be prescribed as a part of treatment for critically ill COVID-19 patients, with possible outcome of saving lives.

2) *Three-Column Format (Enhanced):* In the enhanced setup, an additional question-style instruction was appended to the input sequence along with the claim and evidence. This instruction explicitly guided the model to determine whether the given evidence supports, refutes, or is neutral toward the claim. The three-column format was designed to better structure contextual information and reduce ambiguity, particularly in cases where refutation is implicit rather than explicit.
A representative example formatted under the two-column format is shown below:
Claim: The good news. Properly disinfecting our homes and commonly touched objects helps prevent the spread of all contagious diseases, including COVID-19.
Evidence: As discussed in this review, till effective vaccines and treatments emerge, it is important to understand the scientific rationale of pandemic-mitigation strategies such as wearing facemasks and social distancing, and implement them.
Question: Can vinegar help in preventing COVID-19?

E. Experimental Models

All the experiments were carried out in the Kaggle notebook with the single NVIDIA Tesla T4 (16 GB VRAM) installed. Training each of the transformer-based models (BioBERT, RoBERTa, ClinicalBERT, and SciBERT) to fine-tune took an average of one hour per model when using the chosen hyperparameter settings.

The Meta-Llama-3-8B-Instruct model was significantly more computationally intensive to evaluate. Though it did not do any fine-tuning on the task, zero-shot inference on the entire test split took around 29 hours, with autoregressive token generation and the 8-billion parameter size of the model.

The experimental evaluation was structured into four distinct phases to enable systematic analysis of different modeling strategies:

- Baseline transformer models trained on raw data without class balancing or augmentation
- Transformer models trained with class-weighted loss to address imbalance at the loss level
- An ablation study on BioBERT to isolate the effects of class weighting, augmentation, and input representation
- Zero-shot evaluation using a large language model (LLM) for state-of-the-art comparison

IV. Results and Discussion

This section serves as a structured presentation to enable an easy comparison of the performance of baseline transformer models, imbalance-aware learning strategies, augmentation-based improvements, and large language model performance.

A. Transformer Models on Raw Dataset (Without Class Weights)

Table I shows the performance of transformer-based models trained on the original HealthVer dataset without any class balancing or data augmentation techniques. The evaluation was conducted using BioBERT, RoBERTa, ClinicalBERT, and SciBERT. Performance was measured using class-wise F1-scores, Macro-F1, and overall accuracy.

TABLE I. Performance of Transformer Models on Raw HealthVer Dataset

Model	Evaluation Metrics				
	Refute	***Neutral***	***Support***	***Macro-F1***	***Accuracy***
BioBERT	0.6486	0.8446	0.6844	0.7258	0.7559
RoBERTa	0.6637	0.8446	0.6768	0.7284	0.7284
ClinicalBERT	0.5947	0.8178	0.6300	0.6808	0.7147
SciBERT	0.6307	0.8011	0.6249	0.6855	0.7121

Across all models, relatively good performance is achieved for the Neutral and Support classes, with F1-scores exceeding 0.80 for the Neutral class in most cases. However, performance on the Refute class is consistently lower, with F1-scores ranging between 0.5947 and 0.6637. This discrepancy highlights the challenge of learning refutation patterns from imbalanced data, particularly in the biomedical domain where contradictions are often implicit and semantically nuanced. The results indicate that standard fine-tuning on raw data is insufficient for reliable misinformation detection, especially for minority classes.

B. Transformer Models with Class-Weighted Loss

To mitigate the impact of class imbalance, a class-weighted cross-entropy loss function was applied during training, assigning higher importance to the underrepresented Refute class. The resulting performance is presented in Table II for the same set of transformer models.

TABLE II. Performance of Transformer Models with Class-Weighted Loss

Model	Evaluation Metrics				
	Refute	***Neutral***	***Support***	***Macro-F1***	***Accuracy***
BioBERT	0.6650	0.8342	0.6685	0.7226	0.7512
RoBERTa	0.6637	0.8446	0.6768	0.7284	0.7538
Clinical-BERT	0.5947	0.8178	0.6300	0.6808	0.7147
SciBERT	0.6307	0.8011	0.6249	0.6855	0.7121

The introduction of class weights leads to slight improvements in Refute F1-scores, particularly for BioBERT, where the score improves from 0.6486 to 0.6650. Despite this gain, there is minimal change in Macro-F1 and overall accuracy. These findings suggest that while loss-level reweighting partially improves sensitivity to minority classes, it is insufficient to address deeper semantic underrepresentation. This observation motivates the exploration of data-level strategies that enhance the learning signal for refutation.

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C. Ablation Study on BioBERT

To analyze the individual contribution of different enhancement strategies, an ablation study was conducted using BioBERT as the base model. The results are summarized in Table III, comparing multiple configurations including baseline training, class-weighted loss, paraphrasing-based augmentation, and the combined use of augmentation with the three-column input representation.

TABLE III. ABLATION STUDY ON BIOBERT

Configuration	Evaluation Metrics				
	Refute	***Neutral***	***Support***	***Macro-F1***	***Accuracy***
Baseline	0.6486	0.8446	0.6844	0.7258	0.7559
With Class Weights	0.6650	0.8342	0.6685	0.7226	0.7512
With Augmentation	0.7264	0.8368	0.6926	0.7519	0.7720

The Refute F1-score increases substantially to 0.7264, accompanied by improvements in Macro-F1 and overall accuracy. This improvement indicates that paraphrasing-based augmentation effectively increases exposure to diverse refutation patterns, while the three-column input format provides explicit guidance for the model's decision-making process. The ablation analysis confirms that neither class weighting nor augmentation alone fully explains the observed gains; instead, their combination is essential.

D. Zero-Shot LLM Evaluation

As a reference model, the instruction-tuned Meta-Llama-3-8B-Instruct was evaluated in a zero-shot setting on the HealthVer test split. The corresponding performance is reported in Table IV.

TABLE IV. ZERO-SHOT PERFORMANCE OF META-LLAMA-3-8B-INSTRUCT

Model	Evaluation Metrics				
	Refute	***Neutral***	***Support***	***Macro-F1***	***Accuracy***
Meta-Llama-3-8B Instruct	0.7251	0.8654	0.7589	0.7831	0.8391

Despite the absence of task-specific fine-tuning, the LLM achieves the highest accuracy (0.8391) and Macro-F1 score (0.7831) among all evaluated models. These results demonstrate the effectiveness of LLMs in capturing complex biomedical relationships and underscore their potential for automated health misinformation detection, even in zero-shot settings.

V. CONCLUSION AND FUTURE WORK

In this work, an AI-based method of automated health misinformation detection was proposed through the formulation of the task as the stance classification of medical and policy-related claims into Support, Refute, and Neutral categories. Specifically, the overall performance of the Meta-Llama-3-8B-Instruct model was the highest in a zero-shot task and better than the performance of fine-tuned transformer baselines. BioBERT with Refute-class augmentation and enhanced input representation was among the non-LLM models, and it demonstrated competitive results with significant improvements in Refute-class detection, essential in detecting harmful misinformation.

It can be further studied in future work that reinforcement learning with expert feedback (RLHF) may achieve superior model reasoning and retrieval-augmented generation (RAG) may achieve explicit evidence grounding and minimise hallucinations. Also, the framework can be extended to real-time implementation to social media and health information systems, and large-scale clinical validation to enhance scalability and real-world effect.

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